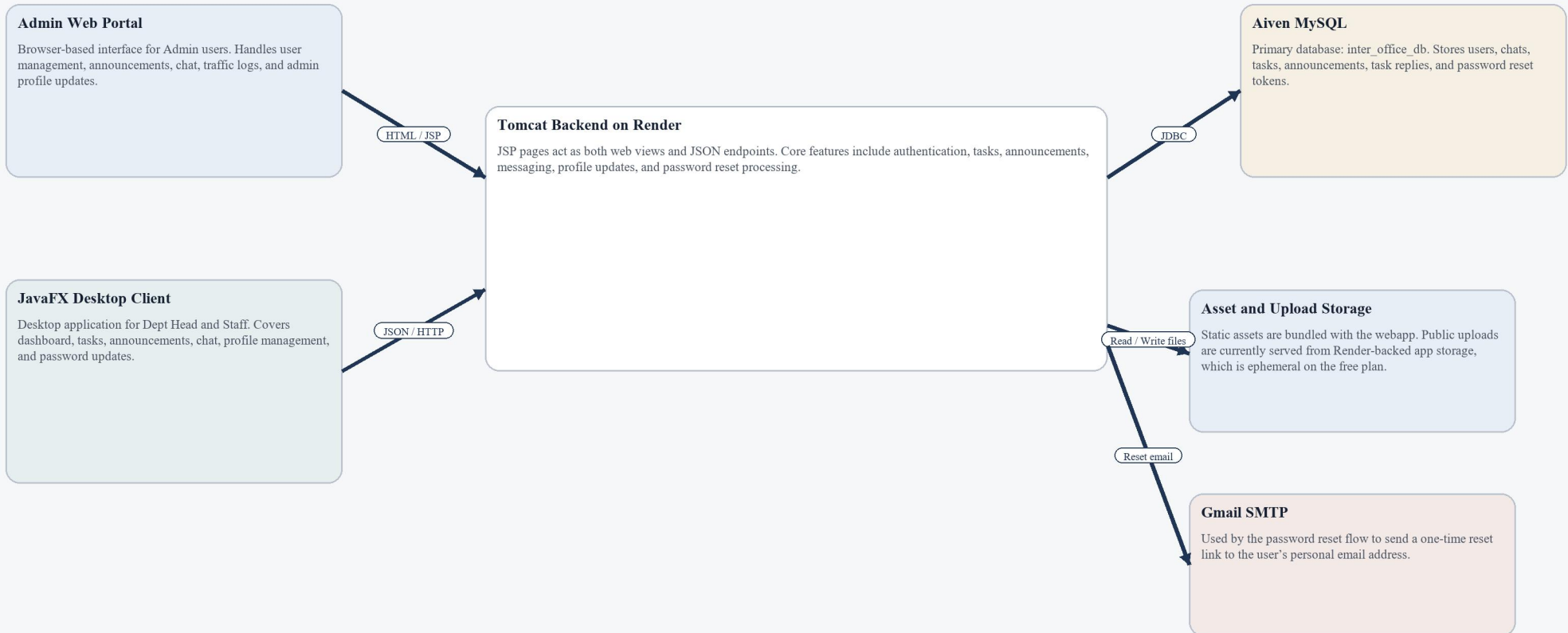




System Architecture

WSU Inter-Office Communication System



Architecture Notes

- Role model: Admin uses the web portal only, while Dept Head and Staff use the desktop client.
- Local development uses Tomcat under `http://localhost:8080/backend-web`, while the public deployment is served from `https://wsu-inter-office-system-backend.onrender.com/`.
- The backend uses direct JDBC from JSPs. This is functional, but tightly couples SQL, transport, and page logic.
- Passwords are upgraded to bcrypt on successful login, and reset links are issued through the backend to personal email addresses.



Architecture Flow View

Request paths, modules, and runtime dependencies

1. Authentication

Desktop LoginController and admin login pages both reach backend auth endpoints. Successful login creates an HTTP session and returns role-linked data.

Calls

2. Core Services

Task, announcement, chat, profile, and password-reset JSP endpoints act as the application service layer currently exposed over HTTP.

SQL

3. Persistence

Each service endpoint talks to MySQL using direct JDBC with repeated connection strings and SQL statements.

Data

4. Output

Responses are either rendered HTML pages for the web portal or JSON payloads for the desktop client.

Desktop-side Modules

MainApp -> Login.fxml / Dashboard.fxml

Controllers: LoginController, DashboardController, ChatController, Task* controllers, ProfileController

Transport: HttpConnector + multipart upload helpers

State: UserSession, model classes, profile image utilities

Navigation: dashboard overview, tasks page, announcements page, profile view, office chat

Backend-side Modules

Public and admin web pages under admin/ plus index.jsp

JSON-style APIs under api/ including auth.jsp, tasks.jsp, announcements.jsp, get_messages.jsp, get_users.jsp

Shared auth/profile helpers such as auth_check.jsp and account_helpers.jspf

Upload storage under assets/uploads and assets/img

Runtime deployment on local Tomcat or public Render root context